

Dr. Prakash Sarkar

Research publication portfolio

Affiliation

Kashi Sahu College, Seraikella, Jharkhand, India

Research areas

cosmology; large-scale structure; the cosmic web

Scholar metrics

426 citations; h-index 9; i10-index 9

Coverage

23 profile records, compiled to 20 distinct works, 2009–2025

Profile

[Google Scholar record](#)

Year-wise publication matrix

Rows are shaded by complete year blocks. Position is the researcher’s place in the abbreviated author list (position/total). A dagger marks a record whose attribution is discussed in the reading notes.

Year	Pos.	Authors (abbrev.)	Paper title	Cited	Journal / publication
2025	3/3	A. Nandi; B. Pandey; P. Sarkar	Tracing correlations between galaxy properties across the Cosmic Web: An IllustrisTNG-based study	3	Astronomy and Computing 55, 101023
2025	6/6	S. K. Pal; S. Dasgupta; A. Datta; S. Majumdar; S. Bag; P. Sarkar	Interpreting the H I 21 cm cosmology maps through Largest Cluster Statistics. Part II. Impact of the realistic foreground and instrumental noise on synthetic SKA1-Low observations	3	Journal of Cosmology and Astroparticle Physics 2025 (10), 096
2024	3/3	A. Nandi; B. Pandey; P. Sarkar	The correlations between galaxy properties in different environments of the cosmic web	9	Journal of Cosmology and Astroparticle Physics 2024 (02), 012
2023	–	S. Dasgupta; S. K. Pal; S. Bag; S. Dutta; S. Majumdar; A. Datta; A. Pathak; et al.	Interpreting the H I 21-cm cosmology maps through Largest Cluster Statistics. Part I. Impact of the synthetic SKA1-Low observations	12	Journal of Cosmology and Astroparticle Physics 2023 (05), 014
2023	1/3	P. Sarkar; B. Pandey; S. Sarkar	The maximum extent of the filaments and sheets in the cosmic web: an analysis of the SDSS DR17	15	Monthly Notices of the Royal Astronomical Society 519 (3), 3227–3236
2022	–	A. Pathak; S. Bag; S. Dasgupta; S. Majumdar; R. Mondal; M. Kamran; et al.	Distinguishing reionization models using the largest cluster statistics of the 21-cm maps	18	Journal of Cosmology and Astroparticle Physics 2022 (11), 027
2021	1/4	P. Sarkar; B. Khirali; U. Laha; P. Sahoo	Exact solution of two-potential system under same range approximation and its implication [†]	18	International Journal of Modern Physics E 30 (08), 2150066
2019	3/6	S. Bag; R. Mondal; P. Sarkar; S. Bharadwaj; T. R. Choudhury; V. Sahni	Studying the morphology of H I isodensity surfaces during reionization using Shapefinders and percolation analysis	40	Monthly Notices of the Royal Astronomical Society 485 (2), 2235–2251
2018	3/5	S. Bag; R. Mondal; P. Sarkar; S. Bharadwaj; V. Sahni	The shape and size distribution of H II regions near the percolation transition	50	Monthly Notices of the Royal Astronomical Society 477 (2), 1984–1992
2017	3/6	J. Bagchi; S. Sankhyayan; P. Sarkar; S. Raychaudhury; J. Jacob; P. Dabhade	Saraswati: an extremely massive ~200 megaparsec scale supercluster	56	The Astrophysical Journal 844 (1), 25

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Year	Pos.	Authors (abbrev.)	Paper title	Cited	Journal / publication
2016	1/5	P. Sarkar; S. Majumdar; B. Pandey; A. Kedia; S. Sarkar	The many scales to cosmic homogeneity: Use of multiple tracers from the SDSS	4	arXiv:1611.07915 (preprint)
2016	4/4	S. Konar; D. Mukherjee; D. Bhattacharya; P. K. Sarkar	Gravitational waves from surface inhomogeneities of neutron stars [†]	5	Physical Review D 94, 104036
2014	1/1	P. Sarkar	Quantifying the Cosmic Web using the Shapefinder diagnostic	–	Proceedings of the International Astronomical Union 11 (S308), 250–253
2014	3/5	S. Sankhyayan; J. Bagchi; P. Sarkar; V. Sahni; J. Jacob	Identification of Extremely Large Scale Structures in SDSS-III	–	Proceedings of the International Astronomical Union
2014	1/3	P. Sarkar; B. Pandey; S. Bharadwaj	The luminosity, colour and morphology dependence of galaxy structures in the Sloan Digital Sky Survey	–	Journal of Physics: Conference Series 484 (1), 012050
2012	1/1	P. Sarkar	Exploring the Cosmic Web in the Sloan Digital Sky Survey Data Release Six using the Local Dimension	–	Journal of Physics: Conference Series 405 (1), 012030
2012	1/3	P. Sarkar; B. Pandey; S. Bharadwaj	Exploring the cosmic web in the Sloan Digital Sky Survey Data Release 7 using the Local Dimension	6	Monthly Notices of the Royal Astronomical Society 423 (1), 955–961; arXiv:1105.4469
2010	1/1	P. Sarkar	Quantifying the scale of Homogeneity and the Cosmic Web using Galaxy Redshift survey	–	IIT Kharagpur
2009	1/4	P. Sarkar; J. Yadav; B. Pandey; S. Bharadwaj	The scale of homogeneity of the galaxy distribution in SDSS DR6	167	Monthly Notices of the Royal Astronomical Society: Letters 399 (1), L128–L131
2009	1/2	P. Sarkar; S. Bharadwaj	The Local Dimension: a method to quantify the Cosmic Web	20	Monthly Notices of the Royal Astronomical Society: Letters 394 (1), L66–L69

Reading notes. Alternating pale-blue and white bands apply to complete profile-year groups. Author position is the researcher’s place in the abbreviated author list (position/total); a dash means the profile truncates the list before his name appears, so his place cannot be read off. The pale shading marks the 2025, 2023, 2021, 2018, 2016, 2012 and 2009 groups.

Deduplication. Of the 23 profile records, three are not separate works and are merged. Two carry the same title and year — *Identification of Extremely Large Scale Structures in SDSS-III* is listed once with a full author list and once with only the lead author — and the 2011 preprint of the SDSS Data Release 7 local-dimension paper is the same work as its 2012 journal version. A 2026 corrigendum to the 2025 *Astronomy and Computing* paper is a correction notice rather than a new study, so it is recorded against that paper instead of as its own row. The result is the 20 distinct works tabulated here.

Affiliation history. The header records the affiliation the profile currently shows. The papers name two earlier ones: IIT Kharagpur for the 2009–2012 work, which includes the 2010 thesis, and the National Institute of Technology, Jamshedpur from 2016 to 2021.

Cross-field records. Two records sit outside the cosmology programme that the rest of this profile forms, and both are retained.[†] “Gravitational waves from surface inhomogeneities of neutron stars” is a full-name match: the paper’s own affiliation line reads “Prakash Sarkar, NIT Jamshedpur”, the institution his 2018 and 2019 cosmic-web papers also name. “Exact solution of two-potential system under same range approximation” is nuclear scattering theory, bylined with initials only — “P. Sarkar”, Department of Physics, NIT Jamshedpur, where all three of his co-authors on that paper are based. The published metadata gives no given name for that byline, so it is the weaker of the two attributions; it is kept because the affiliation matches and because both works appear on his own ORCID record (0000-0002-0445-5411).

Preprints. Two records are arXiv items. The 2016 cosmic-homogeneity study is an arXiv-only preprint, and the 2011 SDSS DR7 paper is the preprint companion to its 2012 journal version, which is why the same work appears once with both identifiers. Titles are reproduced from the profile with capitalization normalised.

Data snapshot. Public Google Scholar profile, checked 27 September 2026. Scholar metrics are dynamic and may change after this snapshot.